

The Jacobian Conjecture after the Three-Dimensional Counterexample: Exact Validation, a General Seed Family, and the Two-Dimensional Obstruction

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Abstract

On 2026-07-19, Levent Alpöge announced (crediting the language model Claude Fable 5 and a question posed by Akhil) an explicit polynomial map $F : \mathbb{C}^3 \rightarrow \mathbb{C}^3$ with constant Jacobian determinant -2 and a repeated value, refuting the Jacobian conjecture for every $N \geq 3$. We report the results of an exact-arithmetic experimental program built on this event. First, we validate the counterexample independently and compute the full fiber over the announced target: it consists of exactly the three announced points. Second, we reverse-engineer the structure of F : it is a weighted skew-product over the invariants $v = xy$, $t = x^2z$, its Keller property reduces to a two-variable identity, and its fibers are governed by a cubic in a single composite variable. Third, we derive and verify a general constructor: every admissible triple (seed polynomial p , scale k , section q) yields a Keller map with determinant $-kp(1)^2$ and generic fiber degree $\deg p + 1$; the announced map is the minimal instance, and the section tail is a generalization axis beyond the announced family. We exhibit new explicit counterexamples with rational collision certificates in fiber degrees 4 and 5. Fourth, we localize why the mechanism cannot reach the still-open two-variable case: a nontrivial $\mathbb{G}_m$ -action on $\mathbb{C}^2$ has a single weight-0 invariant, the equivariant Keller condition collapses to one univariate differential identity, and the two-invariant potential form that generates the family has no room. All computations are exact (rational or symbolic) and reproducible from the repository; every claim below is labeled as machine-verified, derived, or conjectural.

1 Introduction

Let k be a field of characteristic 0. A polynomial map $F = (F_1, \dots, F_N) : k^N \rightarrow k^N$ is a *Keller map* if $\det JF$ is a nonzero constant. The Jacobian conjecture $JC(N)$, stated in two variables by Kraus in 1884 (with a flawed proof whose gap, control of ramification at infinity, remained the subject's central obstacle [6]) and popularized by Keller in 1939 [4], asserted that every Keller map has a polynomial inverse. Over $\mathbb{C}$, invertibility is equivalent to injectivity, so a refutation requires exactly one repeated value. The classical positive ladder includes Wang's degree-2 theorem, Moh's verification of $JC(2)$ up to degree 100, the Bass–Connell–Wright reduction to cubic homogeneous form [2], and Drużkowski's cubic-linear form [3]; Pinchuk [5] refuted the strong real variant by a non-proper local diffeomorphism with non-constant Jacobian. The conjecture is Problem 16 of Smale's list [7].

On 2026-07-19/20, Alpöge announced the map, with $u = 1 + xy$,

$$F = (u^3z + y^2u(4 + 3xy), \quad y + 3xu^2z + 3xy^2(4 + 3xy), \quad 2x - 3x^2y - x^3z), \quad (1)$$

with $\det JF \equiv -2$ and $F(0, 0, -\frac{1}{4}) = F(1, -\frac{3}{2}, \frac{13}{2}) = F(-1, \frac{3}{2}, \frac{13}{2}) = (-\frac{1}{4}, 0, 0)$ [1]. Appending identity coordinates refutes $JC(N)$ for all $N \geq 3$; the case $N = 2$ remains open. At the time of writing no formal paper exists; the announcement is a machine-checkable certificate, which makes an exact experimental program both possible and necessary. This manuscript maps the findings of that program (repository: CAOS.RESEARCH, problem `algebraic-geometry/jacobian-conjecture`; experiments cited as `EXP-nnn`). Each claim is labeled: [MV] machine-verified exact identity, [D] derived analytically and certified on instances, [C] conjecture.

2 Exact validation and the full fiber (EXP-001)

Theorem 2.1 ([MV]). *For F as in (1): $\det JF = -2$ identically; the three announced points are pairwise distinct with the common image $(-\frac{1}{4}, 0, 0)$; and the fiber $F^{-1}(-\frac{1}{4}, 0, 0)$ consists of exactly those three points. In particular F is a non-injective Keller map and $JC(N)$ is false for every $N \geq 3$.*

The fiber computation is a lex Gröbner basis over $\mathbb{Q}$: the eliminant in z is $\frac{1}{8}(2z - 13)(4z + 1)$, and on the fiber $3x + 2y = 0$, $12y^2 = 4z + 1$. All coefficients of (1) are rational, so non-injectivity holds equally over $\mathbb{R}$.

3 The structure of the counterexample (EXP-002)

Theorem 3.1 ([MV]). *F is equivariant for the $\mathbb{G}_m$ -action with source weights $(1, -1, -2)$ and target weights $(-2, -1, 1)$. Writing $v = xy$, $t = x^2z$, $u = 1 + v$, the reduced data*

$$a_1 = x^2P, \quad b_1 = xQ, \quad c_1 = R/x = 2 - 3v - t$$

are polynomials in (v, t) ; each component of F is affine in z with coefficient vector $(u^3, 3xu^2, -x^3)$; and the generating identity $a_1 = \frac{ub_1}{2} - u^2 + \frac{u^3c_1}{2}$ holds.

Lemma 3.2 (Skew-product determinant lemma, [MV] with generic symbolic coefficients). *For any polynomials a_1, b_1, c_1 in (v, t) , the map $(a_1/x^2, b_1/x, xc_1)|_{v=xy, t=x^2z}$ satisfies*

$$\det JF \cdot c_1^2 = -J_2, \quad J_2 = \det \frac{\partial(V', T')}{\partial(v, t)}, \quad V' = b_1c_1, \quad T' = a_1c_1^2.$$

For F , $J_2 = 2c_1^2$ [MV]: the Keller property is a *two-variable* identity; the third dimension only carries weights. Fibers are governed by a single cubic: with $w = uc_1/2$ and $\Phi(w) = w^2 - w^3$,

$$\Phi(w) = \frac{1}{4}(wQR - PR^2) \quad \text{identically [MV]}, \quad (2)$$

so preimages of (A, B, C) correspond to roots of $4\Phi(w) - wBC + AC^2 = 0$; at the announced target this degenerates to $\Phi(w) = 0$ with roots $\{0, 0, 1\}$, matching the three preimages' w -values $(1, 0, 0)$ [MV]. Injectivity dies by *non-properness*: a Keller map is a local diffeomorphism, so sheets merge only at infinity; this is Pinchuk's loophole and the modern face of Kraus's gap.

4 The seed family: a general counterexample constructor (EXP-003/004)

The identification $c_1 = h(v) - t$ with $h = p/w$ is a degree-2 coincidence: an initial constructor built on it is *provably unsolvable* for $\deg p \geq 3$ (EXP-003, an exact unsolvability statement). The failure analysis yields the correct *potential form*.

Theorem 4.1 (Constructor v2, [D], certified on instances [MV]). *Let $p(w) = \sum_{i=1}^d p_i w^i$ satisfy*

$$p(0) = 0, \quad \int_0^1 p = 0, \quad p(1) \neq 0, \quad p'(1) \neq 2p(1),$$

let $k \neq 0$, $m = -kp(1)$, and let $q(v)$ be any polynomial with $q(0) = k$ and $q'(0) = k(p(1) - p'(1))/(p'(1) - 2p(1))$, with arbitrary higher-order tail. Set $\Phi(w) = \int_0^w p$, $s = c_1 = q(v) - t$, $w = us/k$, and define

$$V' = k^2 p(w) + ms, \quad T' = wV' - k^2 \Phi(w), \quad b_1 = V'/s, \quad a_1 = T'/s^2.$$

Then a_1, b_1 are polynomials in (v, t) , and $F_{p,k,q} = (a_1/x^2, b_1/x, xc_1)$ is a polynomial Keller map of $\mathbb{C}^3$ with

$$\det JF_{p,k,q} = -\frac{m^2}{k} = -kp(1)^2 \quad (\text{constant, for every admissible } (p, k, q)),$$

generic fiber degree $d + 1$, and the exact reconstruction inverse: from a root w of the fiber polynomial $k^2 \Phi(w) - (wBC - AC^2)$,

$$s = \frac{BC - k^2 p(w)}{m}, \quad u = \frac{k w}{s}, \quad x = \frac{C}{s}, \quad v = u - 1, \quad t = q(v) - s, \quad y = \frac{v}{x}, \quad z = \frac{t}{x^2}.$$

The announced map (1) is the instance $p = 2w - 3w^2$, $k = 2$, q affine; at $d = 2$ this shape is rigid up to the scale k [MV].

The three admissibility conditions are exactly the low-weight polynomiality constraints at the origin; the potential form makes $J_2 = m^2 s^2 / k$ automatic, so *the Keller property costs nothing*: this is the structural reason the conjecture is false in dimension ≥ 3 .

Theorem 4.2 (New explicit counterexamples, [MV]). *The following maps are non-injective polynomial Keller maps of $\mathbb{C}^3$, each certified by an engineered rational collision:*

Seed p	k	$\det$	Component degrees	Generic fiber degree
$w - 2w^3$	3	-3	(12, 11, 4)	4
$8w - 12w^2 + 4w^3 - 5w^4$	14	-350	(17, 16, 4)	5
$2w - 3w^2$, section tail v^2	2	-2	(8, 7, 5)	3

For the first: $(-\frac{1}{15}, 18, 5130)$ and $(\frac{1}{24}, -18, -10368)$ both map exactly to $(-54, -54, 1)$. Collisions are engineered by prescribing two rational roots $w_1 \neq w_2$ and solving the linear system $w_i \tilde{V} - \tilde{T} = k^2 \Phi(w_i)$ for the target invariants.

The section-tail instance changes the component-degree pattern at fixed fiber degree and determinant: the family is strictly larger than the seed-only family circulating informally. Every negative rational number is realized as a Keller determinant within the family.

5 The two-dimensional obstruction (EXP-005)

Theorem 5.1 ([MV] with generic symbolic f, g). *Let $\mathbb{G}_m$ act on $\mathbb{C}^2$ with weights $(1, -a)$, $a \geq 1$, and let $F = (x^{b_1}f(v), x^{b_2}g(v))$, $v = x^ay$, be an equivariant polynomial map. Then*

$$\det JF = x^{b_1+b_2+a-1} (b_1fg' - b_2f'g),$$

so the Keller condition forces $b_1 + b_2 = 1 - a$ and the single univariate identity $b_1fg' - b_2f'g = \text{const} \neq 0$.

Proposition 5.2 (One-invariant collapse). *If the reduced data (V', T') of a weighted skew-product depend on a single variable, then $dV' \wedge dT' = 0$, hence $J_2 \equiv 0$ and $\det JF \equiv 0$: no Keller map arises. Since a nontrivial $\mathbb{G}_m$ -action on $\mathbb{C}^2$ has one independent weight-0 invariant (against two in $\mathbb{C}^3$), the potential-form mechanism of Theorem 4.1 is structurally unavailable in dimension 2.*

Additionally [MV]: no coordinate slice of (1) (each variable frozen at rational test values, any output pair) is a 2D Keller map.

Conjecture 5.3 (2D equivariant rigidity, [C]). *Every $\mathbb{G}_m$ -equivariant Keller map of $\mathbb{C}^2$ is invertible.*

Our first enumeration scan was vacuous (zero surviving instances at the tested degrees) and is recorded as such; a branch-symbolic enumeration is queued (EXP-006). The honest map of remaining routes to a hypothetical 2D counterexample: no symmetry at all; non-reductive ($\mathbb{G}_a$) structures, which face the same one-invariant collapse in naive form; or gradings at infinity in the Newton-polygon/Abhyankar sense. Conversely, a proof of $JC(2)$ in this program's terms is a properness statement: the reconstruction inverse of Theorem 4.1 shows exactly how asymptotic values are produced in dimension 3 (denominators $BC = k^2p(w)$), and whether one-invariant geometry leaves room for such escape is, in our view, the sharpest transferable open question.

6 Discussion and program

Why the conjecture is false. In the equivariant class, being Keller is a two-variable potential identity, satisfiable for free at any fiber degree ≥ 3 ; nothing in the determinant sees properness.

What is essential. Two independent weight-0 invariants; the potential form; valuation conditions at the origin. **Consequence cascade (UNVERIFIED here).** Via known implications the refutation propagates to the Mathieu conjecture for $SU(3)$, the Gaussian moments conjecture, Zhao's vanishing conjecture, the Image conjecture, and, via stable equivalence, the Dixmier conjecture; primary-source verification of each chain is queued before this program asserts them.

Next experiments. EXP-006 (branch-symbolic 2D enumeration for Conjecture 5.3); EXP-007 (exact asymptotic varieties of F and the new instances); minimality of the degree vector $(7, 6, 4)$; the Bass–Connell–Wright push-through of F to explicit cubic form.

Reproducibility

Every theorem labeled [MV] is an exact sympy computation over $\mathbb{Q}$ with a persisted script and output in the repository (experiments EXP-001 through EXP-005 under `problems/algebraic-geometry/jacobi`; each experiment declares its hypothesis before the run and records an adversarial-validation route with its verdict.

References

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